Probability 1 - Excercises

Week 1 2025

1.1 John, Jim, Jay and Jack have formed a band consisting of 4 instruments.

(a) If each boy can play all 4 instruments, how many different arrangements are possible?

(b) What if John and Jim can play all four instruments, but Jay and Jack can only play piano and drums?

1.2 For years, telephone area codes in the United States and Canada consisted of a sequence of three digits. The first digit was an integer between 2 and 9, the second digit was either 0 or 1, and the third digit was any integer from 1 to 9.

(a) How many area codes were possible?

(b) How many area codes starting with 4 were possible?

1.3 In how many ways can 8 people be seated in a row if

(a) there are no restrictions on the seating arrangement?

(b) persons A and B must sit next to each other?

(c) there are 4 men and 4 women, and no 2 men or 2 women can sit next to each other?

(d) there are 5 men and they must sit next to each other?

(e) there are 4 married couples, and each couple must sit together?

1.4 How many 5-card poker hands are there?

1.5 A dance class consists of 22 students, of which 10 are women, and 12 are men. If 5 men and 5 women are to be chosen and then paired off, how many results are possible?

HW 1.6 (1 point) A committee of 10, consisting of 3 Republicans, 3 Democrats, and 4 Independents, is to be chosen from a larger group of 7 Republicans, 6 Democrats, and 8 Independents. How many committees are possible?

1.7 How many different linear arrangements are there of the letters A,B,C,D,E,F,G for which

(a) A is before B?

(b) A is before B and B is before C?

(c) A and B are next to each other, and C and D are also next to each other?

(d) E is not last in line?

1.8 A bakery has 4 kinds of cakes.

(a) We want to order a tray of 20 pieces. How many different combinations for a tray are there?

(b) We want to order a tray of 20 pieces. What if we insist on having at least one cake of each kind?

(c) If there are only two pieces of each cake left in the bakery and I want to order at least one of each cake, but I have no restriction on the number of cakes that I want to order, how many different cake combinations for a tray are there?

1.9 Consider a 3×5 grid (it means 15 junction points). Suppose that, starting at the bottom left corner $A = (1, 1)$, you can go one step up, or one step to the right at each move. This procedure is continued until the upper right corner $B = (5, 3)$ is reached. How many different paths from A to B are possible?

HW 1.10 (2 points) Similarly to the previous problem, how many different paths are there on a 6×8 grid from $A = (1, 1)$ to $B = (6, 8)$ that go through the point at $(4, 4)$?

HW 1.11 (2 points) (a) If 6 identical blackboards are to be divided among 2 schools, how many divisions are possible? How many, if each school must receive at least 1 blackboard?

(b) What if 6 teachers (definitely not identical) are divided between two schools?

1.12 We put n (distinguishable) balls randomly into n (distinguishable) urns. What is the probability, that exactly one urn remains empty?

1.13 We put 8 rooks (tower) randomly onto a chess board. What is the probability that none of the rooks can knock each other out?

1.14 There are 18 elks living in a forest, 5 of them are marked. If 4 elks are captured at random, what is the probability that of those captured exactly 2 will be marked?

HW 1.15 (2 points) We have 10 mint and 10 lemon flavoured candies. The 20 pieces of candy are distributed randomly between Anne and Barbara, such that each girl gets 10 pieces. What is the probability that

(a) all the mint flavoured candies go to Anne?

(b) both girls get exactly 5 pieces of each kind?

1.16 There are n pairs of shoes in a closet. We chose $2r$ shoes randomly ($2r \leq n$). What is the probability, that

(a) there are no pairs chosen?

(b) there is exactly one pair chosen?

(c) there are exactly two pairs chosen?

1.17 We throw a coin until it lands on the same side twice in a row. Prove that the probability of every n long sequence is 2^{-n}. Write down the probability space for the experiment. What is the probability of the following events?

$$\begin{aligned} A &:= \{\text{The experiment ends after less than 6 throws}\} \\ B &:= \{\text{The experiment ends after an even amount of throws }\} \end{aligned}$$

HW 1.18 (3 points) Ariana, Britney and Cardi, are throwing coins, one-by-one. Cardi starts, then Britney, then Ariana, then Cardi again and so on. They continue, until someone throws heads. The one who throws heads first wins the game.

(a) What is the event space (Ω)?

(b) Write down the following events as subsets of the event space, and calculate their probabilities.

$$A = \{\text{Ariana wins}\}, \quad B = \{\text{Britney wins}\}, \quad (A \cup B)^c$$

1.19 Prove that:

$$\binom{n+m}{r} = \binom{n}{0}\binom{m}{r} + \binom{n}{1}\binom{m}{r-1} + \cdots + \binom{n}{r}\binom{m}{0}.$$

♣ **1.20** (3 points) Consider the following combinatorial identity:

$$\sum_{k=1}^{n} k \cdot \binom{n}{k} = n \cdot 2^{n-1}.$$

Present a combinatorial argument for this identity by considering a set of n people, and determining in two ways the number of possible selections of a non-empty committee of any size, and a chairperson for the committee.

Probability 1 - Practice

Week 2 | 18th Sep 2025

Inclusion-exclusion formula for two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Inclusion-exclusion formula for three events:

$$\mathbb{P}(A \cup B \cup C) \;=\; \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$$

Inclusion-exclusion formula for finitely many events A_i, $i \in I$ (where $|I| < \infty$):

$$\mathbb{P}\left(\bigcup_{i \in I} A_i\right) = \sum_{\emptyset \neq J \subseteq I} (-1)^{|J|+1} \cdot \mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \sum_{k=1}^{|I|} \sum_{J \subseteq I \,:\, |J|=k} (-1)^{k+1} \mathbb{P}\left(\bigcap_{j \in J} A_j\right)$$

2.1 Let A, B and C be three events. Use set theory operations to express the following events:

(a) Out of A, B and C, at least k events occur ($k = 1, 2$).

(b) Out of A, B and C, exactly k events occur ($k = 0, 1, 2$).

2.2 How many different outcomes can the experiment have? Identify the sample space of the experiment:

(a) We throw 3 different coins and 2 identical dice.

(b) We throw 3 identical black dice, and 2 identical white dice.

2.3 What has a higher probability? Throwing a die four times, and rolling 6 at least once, or throwing two dice 24 times, and rolling 6 on both together at least once.

2.4 From a population of n balls in a bucket, we color R to be red, and the rest of them to be blue. Now, let's choose a sample of size N from the population. What is the probability that none of the selected balls are red if

(a) we put them back into the bucket after each draw? (drawing with replacement)

(b) we don't put them back into the bucket after each draw? (drawing without replacement)

(c) Still drawing without replacement. What is the probability of choosing exactly i red balls, for $0 \leq i \leq \min\{N, R\}$ fixed?

2.5 There are 37 pockets on the roulette wheel: 18 black, 18 red and 1 green. We spin the wheel 13 times. What is the probability of rolling each color at least once?

2.6 Santa Claus has 40 dark chocolate bars and 20 white chocolate bars. He distributes the chocolate bars randomly among ten children in a way that each child gets six chocolate bars. What is the probability that each of the ten children get at least one white chocolate bar?

HW **2.7** (2 points) We use a 52 card French card deck to play bridge. There are 4 players (N,E,S,W) and each get exactly 13 cards. What is the probability that N has exactly k aces? (k = 0,1,2,3,4)

♣ **2.8** (3 points) Show that for any A, B events:

$$-\frac{1}{4} \leq \mathbb{P}(A)\mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \frac{1}{4}$$

2.9 (a) Let A and B be two events. If $\mathbb{P}(A) \geq 0.8$ and $\mathbb{P}(B) \geq 0.5$ then prove the following:

$$\mathbb{P}(A \cap B) \geq 0.3$$

(b) Let's prove the following equality for any $A_1, A_2, \dots, A_n$ events:

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) \geq \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_n) - (n-1)$$

HW 2.10 (3 points) (a) We throw a die 6 times. What is the probability, that $1, 2, 3, 4, 5$ and 6 all get thrown?

(b) We throw a dice 13 times. What is the probability that $1, 2, 3, 4, 5$ and 6 all get thrown at least once?

2.11 An urn contains 6 red, 4 blue, and 5 green balls. What is the sample space, if a set of 3 balls is randomly selected without replacement?

(a) What is the probability that each of the balls will be of the same color?

(b) What is the probability that each of the balls will be of different colors?

HW 2.12 (2 points) There are 5 red, 7 white, and 5 blue balls in an urn. We draw 6 without replacement. What is the probability that we draw at least one of each color?

2.13 We are watching a horse race with 7 horses. Let A be the event that $HorseA$ is either first, second or third, while B is the event that $HorseB$ is second. What is the probability of $A \cup B$?

2.14 We deal all cards from a well shuffled French deck (52 cards). What is the probability that

(a) the ace of hearts is the 11th distributed card?

(b) the first distributed ace is the 11th distributed card?

(c) the first 4 cards are of different suits? ($\clubsuit, \spadesuit, \heartsuit, \diamondsuit$)

(d) an ace never follows the king of the same suit directly? (that is eg. $K\clubsuit A\clubsuit$ do not appear in the sequence)

HW 2.15 (3 points) 11 married couples sit down randomly to a round table with 22 chairs around it.

(a) What is the probability that Anne, Betty and Caroline all sit next to their husbands?

(b) What is the probability that no husband sits next to his wife?

2.16 In a certain village, there are 20 families: 5 families with a single child, 7 families with 2, 4 families with 3, 3 families with 4 and 1 family with 5 children.

(a) We choose a family at random. What is the probability that there are i children in this family?

(b) We choose a child at random. What is the probability that the family of this child has i children?

Probability 1 - Practice

Tutorial no. 3 25th Sep 2025

3.1 We put N balls into n boxes, with all n^N possibilities having equal probability. Assuming that a given box is not empty, what is the conditional probability that there are K balls in that box $(1 \leq K \leq N)$?

3.2 We throw three dice. Assuming that the number on each die is different, what is the conditional probability that at least one of them is a 6?

3.3 Die α has 4 red and 2 white sides, while die β has 2 red and 4 white sides. We toss a coin. If the outcome is heads, we use die α, and if the outcome is tails, we use die β. We throw n times with the die chosen.

(a) What is the probability, that the k-th throw lands on red? $(k = 1, 2 \ldots, n)$

(b) Assuming that the first throw was red, what is the conditional probability that the second throw will also be red? $(k = 1, 2 \ldots, n)$

(c) Assuming that the first $k-1$ throws were red, what is the conditional probability that the k-th throw will also be red? $(k = 1, 2 \ldots, n)$

3.4 There are 2 balls in an urn. They were painted to either black or gold with probability $\frac{1}{2} - \frac{1}{2}$ independently.

(a) We have been told that at least one of the balls is gold. What is the probability of both balls being gold?

(b) We pull out a ball and it is golden. What is the probability that the other ball is also golden?

HW **3.5** (3 points) We throw with one blue, one yellow, one green and one red die. Let the three numbers thrown be B, Y, G and R.

(a) What is the probability that all four numbers thrown are different?

(b) Assuming that all four numbers are different, what is the conditional probability of $B < Y < G < R$?

(c) How much is $\mathbb{P}(B < Y < G < R)$?

3.6 Three chefs, A, B and C are trying to bake special cakes. They have 0.02, 0.03 and 0.05 probabilities of overcooking a cake, respectively. A bakes 50% of the cakes, B bakes 30% and C bakes the remaining 20% of cakes. What percent of the ruined cakes did A bake?

3.7 0.1% of human fetuses have the Down syndrome. There is a clinical test that for each fetus with the Down syndrome gives a false negative with probability 1%, while for each fetus without the syndrome gives a false positive with probability 5%. The test tells a couple that their fetus has Down syndrome. What is the probability that this is indeed the case?

♣ **3.8** (3 points) A hunter sees a fox at a distance of 40 meters and shoots at it. If the fox survives, it runs away with a speed of 10 m/s. It takes 5 seconds for the hunter to reload and shoot again, continuously. The fox has a $\frac{1}{4}$ probability of surviving a shot, no matter how many shots have hit him previously. We know that:

$$\mathbb{P}(\text{The hunter hits the fox from a distance of x meters}) = \frac{675}{x^2}$$

Does the fox have a positive probability of surviving?

HW **3.9** (3 points) Shane spends $\frac{4}{5}$ of his day at the pub. There are five bars in the village, and because he has no preference, he picks each pub with an equal probability and spends the day there. We try to find him and visit four bars, but he wasn't in any of them. What is the conditional probability that he is in the fifth pub?

3.10 (Monty Hall problem) Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

HW 3.11 (2+2 points) There are two urns, with red and blue balls. In the first urn, there are 5 red and 4 blue balls, while in the second urn, there are 7 red and 3 blue balls.

(a) Scenario 1: we take a ball from the first urn and put it into the second urn. We then take a ball from the second urn and put it into the first urn. Then, we take another ball from the first urn. What is the probability that this ball is red?

(b) Scenario 2: we take 1-1 ball from the two urns at the same time, and switch them. Then we take a ball from the first urn. What is the probability that this ball is red?

3.12 A private facebook group can only be joined if someone from the group invites you. Originally, the group has two members: Adam and Eve. Occasionally, a random person from the group invites a new member. Someone is part of Adam's "clique", if they are Adam themselves, or if someone from Adam's clique invited them.

(a) What is the probability of Adam's clique having k members when the facebook group has 4 members? ($k = 1, 2, 3$)

(b) Let X be the number of people in Eve's clique when the facebook group has n members. What is the probability of the event $\{X = k\}$ for $k \in \{1, \ldots, n-1\}$?. *Hint:* Use mathematical induction.

3.13 100 passengers are waiting to board an airplane with 100 seats. The first person to board the airplane lost their ticket, and so they sit down randomly in one of the seats. The following people board the plane individually, and try to sit down in the seat written on their ticket. If they see someone already sitting in their seat, they sit down randomly in one of the empty seats. What's the probability that the last person to board the plane ends up sitting in the correct seat?

Probability 1 – Exercises

Tutorial no. 4 2nd October 2025

4.1 A couple are in a team on a gameshow, and are asked a question. Both of them have p probability independently to know the answer. Which strategy is better?

(a) One of them are chosen randomly, and they answer without talking to the other.

(b) Both of them think of an answer, and if they agree they give that answer, otherwise they decide using a coin.

4.2 We toss a coin 3 times. Let A be the event that there is both at least one heads and at least one tails tossed. Let B be the event, that there is at most one tails in the outcomes. Are the two events independent?

4.3 We deal all the 52 cards of a French deck one-by-one. Which is the most likely position to see the second Ace?

4.4 We are throwing a standard die until the second 6. How many times do we have to throw the die most likely?

♣ **4.5** (3 points) Let's pick a random number from the set $\{1, 2, 3, \dots, n\}$ with an equal probability. Let A_p be the event that the chosen number is divisible by p, which is a prime.

(a) Let's show that if $p_1, p_2, \dots, p_k$ are prime numbers and n is divisible by $p_1, p_2, \dots, p_k$, then $A_{p_1}, A_{p_2}, \dots, A_{p_k}$ events are (totally) independent.

(b) Let C_n be the event that the randomly chosen number is a relative prime to n. Prove that

$$\mathbb{P}(C_n) = \prod_{p \text{ prime}, p|n} \left(1 - \frac{1}{p}\right)$$

4.6 We put a knight on a random position on a chess board. What is the expected value of the possible moves from that position?

4.7 In a casino, we play the following game: we bet a number from one to six, then a die is thrown three times. If our number appeared at least once then we get a peso for each appearance of our number, but we have to pay one peso if our number did not appear at all. Is the game fair?

HW **4.8** (2 points) Andrew and Bob are playing a game. There are 5 red and 5 green balls in an urn. They draw 2 balls from the urn. If they're both the same color, Andrew pays Bob 1000 forints. If they're different colors, Bob pays Andrew x forint. How much should x be to make the game fair?

4.9 (St. Petersburg paradox) We toss a coin until it lands on heads. If the n^{th} toss is heads, the player wins 2^n forint. Show that the expected value of the reward is infinite.

(a) Is it worth paying 1 million Ft to play the game?

(b) Is it worth paying 1 million Ft per game if we can play as much as we want, and only have to pay after we decide to stop?

4.10 A student has to fill out a test with 20 questions that can be answered with either "yes" or "no". Let's assume that the student knows the answer to each question independently with probability p. With probability q they think they know the answer, but they don't, and with probability r they know that they don't know the answer ($p + q + r = 1$). If the student knows that they don't know the answer, they randomly write "yes" or "no" with 50-50% probability. What is the probability that they answer at least 19 questions correctly?

4.11 Among five players, A, B, C, D, E, the numbers from 1 to 5 are distributed randomly, without repetition. First, A and B compete: whoever has the higher number moves on to play against the next opponent. The one who advances in this way will now face C, then the one who advances from among them will compete with D, then the winner battles with E. Let X be the number of rounds that A wins. Determine the distribution and expected value of X.

HW 4.12 (3 points) A class of 7 women and 4 men wrote a test. Let's assume that there are no ties in their test scores, and order them from best to worst, with each of the orderings having equal probability. Let X denote the placing of the highest ranking woman (eg. $X = 1$ means that the best test was written by a woman). Determine the distribution and the expected value of X.

4.13 Show that for a non-negative, integer valued random variable N, where $\mathbb{E}(N) < \infty$

$$\mathbb{E}(N) = \sum_{i=1}^{\infty} P(N \geq i)$$

4.14 There are 5 players; A,B,C,D,E and we give each of them a number between 1 and 5, without repetition (they can't have the same number). First, A and B fight, and whoever has the higher number goes onto the next round. Then, the person who won the previous round fights with C, and the person with the higher number wins. Similarly with D and E. Let X be the number of rounds A wins. What is the distribution and expected value of X?

4.15 On 4 buses 148 students are travelling all together. The number of students on each bus is 40, 33, 25 and 50, respectively. Choose a student uniformly at random, and let X denote the number of students on their bus. Also choose a bus driver uniformly at random, and let Y denote the number of students on their bus.

(a) Which is larger, $\mathbb{E}(X)$ or $\mathbb{E}(Y)$?

(b) Calculate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

(c) Calculate $\mathbb{D}^2(X)$ and $\mathbb{D}^2(Y)$.

4.16 Anna and Bianca have two regular dice, one white and one yellow, and they play according to the following rules. First Anna rolls the two dice, and if the two dice have the same number on them, she wins. If Anna rolls different numbers then Bianca gets the dice and now she rolls: if she manages to roll at least one six, she wins. If not then there is no winner in the first round and they repeat the same process, so they roll alternately until one of them wins.

(a) Denote by X the total number of rounds that they play. What is the distribution and the expected value of X?

(b) What is the probability that the entire game ends with Anna winning?

HW 4.17 (3 points) We have two dice, and we're only focusing on the sum. We keep throwing until the sum of the two dice is either 6 or 9.

(a) What is the probability that we stop on 6?

(b) Prove that the number of throws and the sum when we stop are independent. In other words, show that if

$$A_k = \{\text{We throw k times}\} \qquad \text{and} \qquad B = \{\text{We stop on 6}\}$$

then for every $k \geq 1$, A_k and B are independent events.

4.18 There are $m + 1$ urns with m balls in each. In the i-th urn, there are $i - 1$ red and $m - i + 1$ blue balls ($i = 1, \ldots, m+1$). First, we choose an urn uniformly randomly, then we draw n balls from the chosen urn with replacement.

(a) What is the probability that we only draw red balls?

(b) Assuming that the first n draws were red, what is the probability that the $n + 1$-th draw will be red as well?

(c) Calculate the limit of the probability obtained in part (b) for fixed n as $m \to \infty$.

Probability 1 – Exercises

Tutorial no. 5 9th Oct 2025

5.1 Assuming that the ratio of left-handed people is 1% on average, what is the probability that out of 200 randomly selected people, at least 4 are left-handed?

HW 5.2 (2 points) I operate the giant dough mixing machine at the muffin factory. On average, how many raisins need to be in a muffin, if we want to make sure that for a randomly selected muffin, the probability of it containing at least 1 raisin is at least 99%?

5.3 What is the expected value and the variance of a die roll?

5.4 We're reading Glamour Magazine, and we notice that there are around 0.2 typos each page, on average. What is the probability that the next page has (a) 0 typos (b) 2 or more typos?

5.5 At a horse race, the horses run laps and there are many obstacles that they have to jump over. My horse, "Lucky Star", has an equally small probability to knock each obstacle down, independently of the other obstacles, and the probability that she does a lap perfectly is 0.1.

(a) What is the probability that she knocks down at least 4 obstacles in one lap?

(b) At each mistake she gets an injury with probability 1/2, independently of what happens at any other obstacle. What is the probability that she finishes a lap without any injuries?

5.6 A Marathon is held next to the Danube Bend. Sadly, the route was led through a section of tall grass, which was full of ticks. We later found out, that 300 people found one tick on their skin, and 75 found two. Based on this, approximate how many people ran in the race, overall.

HW 5.7 (3 points) In Myocardistan, the rate of heart attacks is 1 per $10,000$ people every month. Let's examine the city of New Cholesterol in Myocardistan, which has $30,000$ inhabitants.

(a) What is the probability that at least 4 people will have a heart attack in a given month?

(b) What is the probability that in a given year, there will be at least 1 month with at least 4 heart attacks?

(c) Starting now, what is the probability that the first month with at least 4 heart attacks will be i months from now? ($i \geq 1$)

5.8 We are rolling a die until getting the second 5.

(a) What is the expected number of rolls we need?

(b) What is the probability that we need more rolls than this expectation?

5.9 We have 10 red, 20 blue and 40 green balls in an urn. We randomly select 7. What is the distribution of the number of blue balls selected? What is their expected number?

5.10 We throw 10 times with a fair coin. Let X be the number of sequences with the same result (For example if $HHTTTTHTHH$ then $X = 5$). What is the distribution of X?

5.11 Let $\mathbb{E}(X) = 1$ and $\mathbb{D}^2(X) = 5$. Calculate the following:

(a) $\mathbb{E}((2 + X)^2)$

(b) $(\mathbb{E}(2 + X))^2$

(c) $\mathbb{D}^2(4 + 3X)$

5.12 We have two dice, one red and one green.

(a) What is the expected value of the minimum of the two numbers, and the maximum of the two numbers?

(b) We threw with both dice 3 times. I only looked at the red one, but I saw that it was 3 each time. What is the probability, that it was bigger than the green one at least once?

HW 5.13 (2 points) We throw three times with a biased coin which gives heads with probability 3/4. Let U be the number of times we throw the previous result again (so U can be $0, 1$ or 2. For example, if HHH then $U = 2$, if HTH then $U = 0$). Calculate the expected value and variance of U!

HW 5.14 (3 points) We throw 10 times with a biased coin that has $\frac{2}{3}$ probability of heads. After each throw that lands on tails, we have to pay $200Ft$, however, we get $300Ft$ after each throw that lands on heads. Calculate the expected value and variance of our winnings at the end of the game.

5.15 Let N be a non-negative, integer valued random variable with $\mathbb{E}(N^3) < \infty$.

Show that:

$$\sum_{i=1}^{\infty} i^2 \mathbb{P}(N \geq i) = \frac{\mathbb{E}(N^3)}{3} + \frac{\mathbb{E}(N^2)}{2} + \frac{\mathbb{E}(N)}{6}$$

♣ **5.16** (3 points) Two people with the same abilities are running a race, on an infinitely long track. Both of their lanes have the same obstacles (ditches), and they both have 2^{-L} probability to jump over a ditch, where L is the width of the ditch, independently of each other or the previous ditches. If one of them falls in a ditch, then the race is over. If both of them fall in the same ditch, then it is a tie, and we start over.

(a) Show that the probability of a tie is less than ε exactly if $L < \log_2(\frac{1+\varepsilon}{1-\varepsilon})$

(b) How big should L be, if we want to minimize the expected number of jumps until someone wins?

5.17 In the center of London, the number of traffic accidents follows a Poisson distribution with parameter 20 on rainy days and parameter 10 on sunny days. In early Novermber in London, the probability that it rains (the whole day) is 0.6; the probability that the weather is clear is thus 0.4. According to The Times, 17 car accidents happened in London's center last Thursday. What is the probability that it was a rainy day?

Probability 1 - Exercises

Tutorial no. 6 16th Oct 2025

6.1 Show that if $X \sim \text{POI}(\lambda)$ and $Y \sim \text{POI}(\mu)$ and if X and Y are independent and if $Z = X + Y$ then $Z \sim \text{POI}(\lambda + \mu)$.

HW 6.2 (2 points) Leo, when he goes hiking, has a small, equal and independent probability to fall with each step and hit his knee, or fall and hit his elbow. On a 10 kilometer hike, on average, he hits his knee three times, and his elbow two times. At most, how long of a hike can his mother let him go on if she wants a $\frac{2}{3}$ probability for him to not hit his knee or elbow?

6.3 The police frequently checks the speed of cars on the highway ("radaring"). Experience shows that if they turn on the radar for five minutes then the probability of detecting at least one speeding car is the same as the probability of not detecting any.

(a) What is the probability that after 20 minutes of radaring, (i) exactly 2, (ii) at least 2 speeding cars will be caught?

(b) How long should the police radar for if they want to catch at least 1 speeding car with a 95% probability?

6.4 Let's assume that the number of gold ores in a given area has $Poi(10)$ distribution. Each gold ore, independently of the others, is discovered with probability $\frac{1}{50}$. Calculate the probability that

(a) exactly 1,

(b) at least 1, and

(c) at most 1 ore is discovered this time.

Hint: First show that if a random variable has $Poi(\lambda)$ distribution, and each event is counted with p probability, independently of each other, then the number of events counted is $Poi(\lambda \cdot p)$.

HW 6.5 (2 points) The average density of trees in a forest is 18 trees in $100m^2$. The trunk of the trees are perfect cylinders, with a diameter of $20cm$ each. We shoot a gun $120m$ away from the edge of the forest, without aiming in the direction of the edge of the forest. What is the probability that we hit a tree? (Note: Disregard the fact that the centre of two trees can't be closer than 20cm to each other).

6.6 Ulysses wants to go home to Ithaca, but he angered Poseidon, who put the ocean full of whirlpools. The density of whirlpools is 2 whirlpools in every 10 square kilometers. If a boat gets closer than $150m$ to a whirlpool, it sinks. Otherwise, the ship remains unharmed. Odysseus doesn't know where the whirlpools are, he just goes straight to Ithaca. He studied Probability 1 at BME, so he knows that he has a 1% probability to make it home safely. How far away is Odysseus from Ithaca?

6.7 Milka introduced a new type of chocolate bar that contains raisins and peanuts. Each chocolate bar consists of 15 squares. Each chocolate bar has 30 raisins in it on average, and the probability that a square of chocolate has no peanuts is $\frac{1}{2}$.

(a) What is the distribution of the peanuts in the chocolate bar?

(b) We break off a piece that consists of two squares. What is the probability that the combined number of peanuts and raisins in this piece of chocolate is less than 2?

6.8 We throw with a fair die until each number from 1 to 6 has occurs at least once. What is the expected value of the number of throws that we need for this?

HW 6.9 (2 points) There are 5 red and 5 blue balls in an urn. We pick 4 balls at random out of the urn. If we picked 2 red and 2 blue, we stop, otherwise we put them back. We continue until we manage to pull out exactly 2 red and 2 blue. What is the probability that we pull exactly n times?

6.10 Amy and Bernadette play the following game. Amy throws with a fair die until the second occurrence of the number 6. She does not tell Bernadette which throw she managed to throw 6 for the first time, but she does tells her which throw she managed to land on 6 for the second time. After this, Bernadette needs to guess when the first 6 could have occurred. What should she guess? If she is smart, with what probability that she guesses correctly?

♣ **6.11** (3 points) Let X be a Poisson distribution with parameter λ. Show that

$$\mathbb{P}(X \text{ is even}) = \frac{1}{2}(1 + \exp(-2\lambda)))$$

6.12 Andrew and Bella are playing the following game. They throw 2 fair dice, and Andrew pays Bella the same amount of forint as the square of the difference of the two numbers, while Bella pays Andrew the sum of the two numbers. Who is favoured in this game?

HW 6.13 (1+2+1 points) Two people are competing in archery. The contestants have a p_1 and p_2 probability to hit the target ($p_1 < p_2$) with each shot. The person who is less skilled starts, and then they alternate. The first person to hit the target wins.

(a) What is the probability that the more skilled person wins?

(b) What is the expected time of the game if there is one shot every minute?

(c) Give a simple expression for the expected time if $p_1 = p_2$. Check that the formula in part (b) gives the same result when $p_1 = p_2$.

6.14 (2 points) The municipality of Randomburg provides free wifi for the citizens. Each wifi hotspot has $50m$ range, and they placed 3 hotsposts per square kilometer randomly. At the very moment when Bob left his house, the university sent an email to all the students to notify them that all teaching activity is cancelled for today due to a bomb alarm. Bob goes to university on a straight path, and he has 70% chance to receive the notification via the free wifi on the way. How far away is Bob's house from the university?

Probability 1 – Exercises

Tutorial no. 7 — 30th Oct 2025

7.1 Which constants A and B are needed for the following to be a Cumulative Distribution Function (CDF):

HW (a) (1 point)

$$F(x) = A \cdot \arctan(x) + B$$

(b)

$$F(x) = \exp(-Ae^{-Bx})$$

HW (c) (1 point)

$$F(x) = \begin{cases} 0, & -\infty < x \leq 0 \\ \frac{A+x}{1+Bx}, & 0 < x < \infty \end{cases}$$

7.2 Which constants C are needed for the following to be a Probability Density Function (PDF)? For these cases, calculate the expected value, and the probability that the corresponding random variable is greater than 2.

HW (a) (1 point)

$$f(x) = \begin{cases} C \cdot \cos(\pi x), & \frac{-1}{2} \leq x \leq \frac{1}{2} \\ 0, & \text{else} \end{cases}$$

(b)

$$f(x) = \begin{cases} C \cdot x^{-5}, & 1 \leq x \\ 0, & \text{else} \end{cases}$$

HW (c) (1 point)

$$f(x) = \begin{cases} C \cdot (-\ln(\frac{x}{3})), & 0 < x \leq 3 \\ 0, & \text{else} \end{cases}$$

7.3 A gas station gets their shipment of oil once a week. Their weekly sales (measured in 1000 liters) is a random variable with PDF

$$f(x) = \begin{cases} 5 \cdot (1-x)^4, & 0 < x < 1 \\ 0, & \text{else.} \end{cases}$$

Then how big of a shipment do they need weekly if they want a less than 0.01 probability to run out of oil in a given week?

7.4 If X is uniform on $[0,1]$, then what is the probability that X, $1-X$ and $\frac{1}{2}$ form a triangle?

7.5 The buses arrive every hour at :00, :15, :30 and :45. If we arrive at random between 7:00 and 7:30, then what is the probability that

(a) I have to wait less than 4 minutes?

(b) I have to wait more than 7 minutes?

(c) What are the answers to a) and b), if I arrive between 7:08 and 7:38 instead?

(d) What are the answers to a) and b), if I arrive between 7:00 and 7:25 instead?

HW 7.6 (3 points) Trains go towards city A every 15 minutes, starting from 7:00am, while they go towards city B every 15 minutes, starting from 7:05am.

(a) If a passenger arrives at the station uniformly between 7:00am and 8:00am, and board whichever train leaves earlier, then what is the probability that they end up going to B? (Or A?)

(b) What if the passenger arrives between 7:10am and 8:10am?

(c) What if the train that goes towards city B leaves every 10 minutes starting from 7:05am?

7.7 A bus goes between city A and city B, which are 100km from each other. The bus breaks down with equal probability (uniform) on the route. If it breaks down, then they call the closest repair center. Currently, the repair centers are in the 2 cities, and one halfway in between them. It has been suggested, that it would be better to place the repair centers at 25km, 50km and 75km. Do we agree? Why? What would be an even better placement? How do we define "better"?

7.8 We choose two points at random on the unit circle. What is the CDF of their distance?

7.9 Let us choose a point inside the three-dimensional unit sphere with a uniform distribution, and let D denote the distance of this point from the center of the sphere. Determine the distribution function and expected value of D.

HW 7.10 (3 points) Let's say we buy a pretzel stick with unit length, which has a piece of salt at s, $(0 < s < 1)$. The pretzel breaks into 2 pieces at a uniform random point in our backpack while we go home from the store. What is the expected value of the length of the piece which has the salt on it?

7.11 Choose a point uniformly inside of a regular hexagon with sides of unit length. Let ξ be the distance of the point from the closest side. What is the CDF and PDF of ξ?

♣ **7.12** (3 points) If I arrive at a meeting $s \in \mathbb{R}$ minutes earlier, I have to pay $a \cdot s$ coins. If I'm late s minutes, I have to pay $b \cdot s$ coins. The amount of time it takes me to arrive is random due to the chaotic public transport, but it has a PDF of $f(x)$. How many minutes before the meeting should I leave, if I want to minimise the expected cost? (Hint: write the integral form and differentiate it).

Probability 1 – Exercises

Tutorial no. 8 6th Nov 2025

8.1 Most IQ tests follow a Normal Distribution with a mean of 100 points and a standard deviation of 15 points. If we believe that these tests are accurate, then

(a) What percentage of people have an IQ between 95 and 110?

(b) How big of an interval around 100 points does 50% of the population fall into?

(c) In a town with 2500 people, on average, how many people will have at least 125 IQ?

HW 8.2 (2 points) Let's say we still believe that these IQ tests (see the previous exercise) are accurate. The test results are divided into three categories; low, medium and high IQ. Out of the participants, 25%, 65% and 10% fall into these categories, respectively. Where should we draw the boundaries between these categories?

8.3 Approximate the probability that out of 100000 poker hands dealt, the number of full houses is between 128 and 158. (In a poker hand of 5 cards, a full house consists of 3 cards of one height, and 2 cards of another height.)

8.4 In Budapest, p percent of people smoke, but we don't know how many. We want to guess their number, so we ask n people, and if k people say they smoke, we guess that p is approximately $p' = \frac{k}{n}$. How many people should we ask if we want to be closer than 0.02 to the truth with probability at least 93%, no matter what the truth is? In other words, how big should we choose n if we want

$$\mathbb{P}(|p' - p| \leq 0.02) \geq 0.93.$$

HW 8.5 (2 points) A factory produces two kinds of coins: fair ones and biased ones; a biased one lands on heads with probability 54%. We have a coin from this factory, but do not know which kind. To decide, we perform the following statistical test: we throw the coin 1000 times, and if it lands on heads at least 520 times, we decide it is biased; if it lands on heads less than 520 times, we consider it to be fair. What is the probability that we are wrong in the case when the coin is fair? And if the coin is biased? *Instruction:* Use the de Moivre-Laplace theorem!

8.6 The time between two metros in Budapest follows an Exponential distribution with an expected time of 2 minutes. I just missed a metro:

(a) What's the probability, that I have to wait at least 5 minutes for the next metro?

(b) I've been waiting for 3 minutes. What's the probability that I have to wait for at least 5 more?

8.7 The expected lifetime of a radioactive isotope, called probabilitium, is 1 year. Assuming that its lifetime follows an exponential distribution, what is its half-life?

8.8 Selena is watching shooting stars in the sky. On a typical summer night, many meteors reach the earth, and Selena sees each of them with a small probability, independently of the others. Typically, she sees around 3 in half an hour. She starts watching the sky on August 19th, 22:00.

(a) What is the probability that she doesn't see a single shooting star between 22:00 and 22:25?

(b) Let T be the time in minutes that Selena needs to wait to see the first shooting star. Calculate the CDF and PDF of T?

(c) What is the lesson?

8.9 The lifetime of the "Mehr Licht" light bulb is exponentially distributed. According to the manufacturer's measurements, 90 percent of light bulbs last for at least one year. How long should the official warranty period of a light bulb be if the company wants no more than 1 percent of customers to complain?

HW 8.10 (3 points) A random glass in the university cafeteria breaks at an exponentially distributed time, with mean 48 months. What is the probability that:

(a) Out of 6 glasses, at most 3 breaks within two years?

(b) Out of 500 glasses, at most 210 breaks within two years? *Instruction:* Use the de Moivre-Laplace theorem!

♣ **8.11** (3 points) In Monte Carlo integration, we are given a continuous function $f : [0,1] \to [0,1]$, and we would like to estimate the value of $\int_0^1 f(x)\mathrm{d}x$. If we have independent $Uni[0,1]$ variables U and V, then (U,V) is a uniform random point in the unit square $[0,1]^2$. Generate n independent samples (U_i, V_i), $i = 1, \dots, n$, and let X_n denote the number of pairs that fall below the graph of f, i.e., for which $V_i < f(U_i)$. Then, the integral is estimated by X_n/n. How large n should be, if we want to get an error larger than 0.1 only with a probability at most 0.02, if the function is

(a) $f(x) = x^3$;

(b) f is not known?

HW 8.12 (3 points) In Budapest, 43% of the citizens would ban smoking in public places. Approximate the probability that out of n people at least 40% of them supports this ban if

(a) $n = 11$,

(b) $n = 101$,

(c) $n = 1001$.

(d) How many people should we ask to guarantee that at least 40% of them will support the ban with probability 95%?

$\Phi(z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Probability 1 – Exercises

Tutorial no. 9 — 13th Nov 2025

9.1 Assume that Z is such a random variable that $2Z$ has the same distribution. What is the distribution of Z?

9.2 Let X be a continuous random variable, with CDF F. What is the distribution of $Y = F(X)$?

9.3 Calculate the PDF of the following random variables:

(a) If ξ is standard normal then what is the PDF of $X := \xi^2$?

HW (b) (1 point) If $\xi \sim \text{EXP}(\lambda)$, then what is the PDF of $X := 3\xi - 2$, and $Z := \sqrt[4]{\xi}$?

(c) If ξ is uniform on the $[-1; 1]$ interval, then what is the PDF of $X := \xi^4$ and $U := \sin(\pi\xi)$?

(d) If ξ is exponential with parameter λ, then what is the PDF of $Y := e^{\xi}$?

HW (e) (2 points) If ξ is uniform on $[-3; 4]$, then what is the PDF of $Z := \xi^2$?

9.4 Let X be a standard Cauchy random variable. (Recall that its PDF is $f(x) = \frac{1}{\pi(1+x^2)}$ for $x \in (-\infty, \infty)$.) Show that X and $1/X$ have the same distribution.

♣ **9.5** (3 point) Let X be a random variable with PDF $\frac{1}{\ln 2}\frac{1}{1+x}$ on the $[0, 1]$ interval, and 0 otherwise. What is the PDF of the fractional part of $\frac{1}{X}$?

9.6 Let ξ denote the distance of the point X and the point of the plane with coordinates $(1; 1)$. Find the probabilty density function of ξ if

(a) X is uniformly distributed on the $[0, 1]$ interval of the x-axis.

(b) X is uniformly distributed on the $[0, 2]$ interval of the x-axis.

9.7 A stick of length ℓ is broken at two independently and uniformly distributed points. What is the expected value of the length of the shortest one out of the three pieces obtained in this way?

9.8 We roll two independent fair dice. What is the joint probability weight function of X and Y if

HW (a) (2 points) X is the minimum of the two throws, Y is their sum;

(b) X is the first throw, Y is the maximum of the two throws?

9.9 From the Leaning Tower of Pisa, I drop a ball from height 40.5 meters. As we know from physics, in t seconds it falls $gt^2/2$ meters, where $g = 9m/s^2$, hence it reaches the ground in exactly 3 seconds. Each of my two friends, Galileo and Isaac, takes a photo of the experiment, at independent random times with distribution Uni$[0; 3]$ sec.

(a) In expectation, how high is the ball on Isaac's photo?

(b) In expectation, how high is the ball on the photo that is taken later?

HW 9.10 (3 points) Jack arrives to school with a delay of $X \sim \text{Uni}[0, 12]$ minutes. Jill arrives with a delay of $Y \sim \text{Uni}[0, 8]$ minutes, independently of Jack.

(a) What is the probability that Jill arrives earlier than Jack?

(b) What is the expectation of the difference $|X - Y|$?

9.11 Let (X, Y) be the pair of coordinates of uniformly chosen point of the unit disc of the plane. Determine the p.d.f. of the marginal distributions.

9.12 Let (X, Y) denote a pair of random variables with joint p.d.f. $f(x, y)$. Are X and Y independent?

(a) $f(x, y) = xe^{-(x+y)}\mathbb{1}[x > 0, y > 0]$.

(b) $f(x, y) = 2 \cdot \mathbb{1}[0 < x < y < 1]$.

9.13 Let X and Y be random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{4}{5}(x + xy + y) & \text{if } 0 < x, y < 1 \\ 0 & \text{else.} \end{cases}$$

Calculate the marginals. Are the two variables independent?

9.14 Calculate the marginal distributions! Are X and Y independent?

HW (a) (2 points) Let the joint probability density function of X and Y be

$$f(x, y) = \begin{cases} A \cdot (x^2 y + y^2 x) & \text{if } 0 < x, y < 1 \\ 0 & \text{else} \end{cases}$$

where A is a positive constant that must be calculated first.

(b) Same question when (X, Y) is uniform on the unit circle.

(c) (X, Y) is a uniform distribution on $D = \{(x, y) \in \mathbb{R}^2 : \; 2x^2 + y^2/2 \leq 1\}$.

9.15 A rectangle has sides with length 1 and a. We choose one point on each side of length 1, with independent uniform distributions. Let X be the distance of these two points. What is the PDF of X?

Probability 1 – Exercises

Tutorial no. 10 — 20th Nov 2025

9.10 Let X and Y be random variables with joint probability density function

$$f(x,y) = \begin{cases} \frac{4}{5}(x + xy + y), & \text{if } 0 < x, y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the marginals. Are the two variables independent?

10.1 *(Buffon's needle problem)* Suppose we have a floor made of parallel strips of wood, each the same width t, and we drop a needle of length l onto the floor. Suppose that the length of the needle is shorter than the width of the strips. What is the probability that the needle will lie across a line between two strips?

10.2 Let $X_1, X_2, \dots \sim \text{Exp}(\lambda)$ be independent exponential random variables.

(a) What is the distribution of $Z = X_1 + X_2$?

(b) What is the probability that $X_1 + X_2 + X_3 > t$?

(c) What is the expectation and variance of $X_1 + X_2 + X_3$?

HW 10.3 (3 points) $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ are two independent exponential random variables. Assume $\lambda_1 \neq \lambda_2$. What is the p.d.f. of $Z = X_1 + X_2$?

10.4 Let the joint PDF of X and Y be

$$f(x,y) = \begin{cases} \frac{1}{y} \exp(-(y + x/y)) & \text{, if } x > 0, y > 0 \\ 0 & \text{, else.} \end{cases}$$

(a) What is the marginal distribution of Y? How much is $\mathbb{E}(Y)$?

(b) Note that calculating the marginal distribution of X explicitely is problematic. Without doing that, calculate $\mathbb{E}(X)$.

(c) What is the conditional density of X, given $Y = y$?

HW 10.5 (3 points) Let the joint density of (X, Y) be $f(x,y) = \frac{3}{2} xy \mathbf{1}_{\{0<x,\, 0<y,\, x+y<2\}}$.

(a) What is the marginal density of X?

(b) What is the conditional density of Y given $X = x$?

10.6 N people arrive for a business dinner. Whenever someone arrives, they look around, and if they see one of their friends, they sit down to one of their tables, otherwise, they sit to a new, empty table. Each person is friends with any other person with p probability, independently from each other. Calculate the expected value of the number of occupied tables.

10.7 10 married couples are seated around a round table, and they all sit down randomly. Let X be the number of couples that manage to sit down next to each other. What is the expected value and variance of X?

10.8 (a) I go to work each day via train and then bus. First, I get on a train, which arrives at the train station at 7:30am, according to schedule. I have 2 minutes to walk to the bus station, where our bus leaves at 7:37am, according to schedule. However, the actual times when the train arrives and the bus leaves are random. More precisely, the train arrives according to a Normal Distribution with a mean of 7:30am and standard deviation of 3, while the time when the bus leaves is normal with mean 7.37am and standard deviation of 4. What is the probability that I only miss the bus (at most) once in a given week (which consists of 5 work days)?

(b) In the afternoon, my bus arrives back to the station at 17:40 and my train leaves at 17:49 according to schedule, with the same standard deviations as above. What is the probability that in the 220 workdays of the year, I will miss the afternoon connection more often than my morning connection?

10.9 On a cold foggy evening, starting at 7 pm, I'm waiting for my girlfriend's bus to arrive. These buses come according to a Poisson process with intensity 1 bus per 10 minutes. Already, the second bus arrives at 7:22 pm, but my girlfriend is not on the bus. At this point, she suddenly appears from the fog, saying that she arrived with the previous bus, we just didn't see each other. Not remembering when the first bus actually came, what is the distribution of the time we spent standing next to each other in the fog?

10.10 Let $X \sim \mathcal{N}(0,1), Y \sim \mathcal{N}(0,2)$ be independent from each other. Let M be a randomly chosen point on the plane with coordinates (X, Y). What is the probability that

(a) $M \in \{(x,y) \in \mathbb{R}^2 : 0 \le x \le 2, |y| \le 1\}$;

(b) $M \in \{(x,y) \in \mathbb{R}^2 : x + y < 3\}$;

HW (c) (2 points) $M \in \{(x,y) \in \mathbb{R}^2 : 1 \le \sqrt{2}|x| + |y| \le 3\}$ (Hint: Rotate the domain);

HW (d) (2 points) $M \in \{(x,y) \in \mathbb{R}^2 : 1 \le 2x^2 + y^2 \le 3\}$ (Hint: Polar coordinates);

♣ (e) (3 points) $M \in \{(x,y) \in \mathbb{R}^2 : x + y \le 0, -2 \le y, |x| \le 1\}$?

10.11 We break an L long stick into two parts, by breaking it at a uniformly chosen point.

(a) Let X be the squared sum of the length of the two parts. What is the PDF of X?

(b) Let Y be the product of the length of the two parts. What is the PDF of Y?

Probability 1 – Exercises

Tutorial no. 11 — 27th Nov 2025

11.1 The random variables X and Y have a jointly absolutely continuous distribution, the marginal density function of X is $f_X(x) = \lambda^2 x e^{-\lambda x}$ if $x > 0$ and 0 if $x < 0$. The conditional distribution of Y given the value of X is uniform on the interval $[0, X]$. Calculate

(a) the joint probability density function of X and Y,

(b) the marginal density function of Y and the expectation $\mathbb{E}(Y)$,

(c) $\mathbb{E}(Y \mid X = x)$ and $\mathbb{E}(X \mid Y = y)$,

(d) $\mathbb{E}(X)$ (at this point it is simpler if we use the previous sub-exercise), and

(e) $\mathrm{Cov}(X, Y)$.

11.2 We throw a coin 10 times, with a probability p of heads and probability $(1 - p)$ of tails at each throw. Let X be the number of *pure runs* (e.g. in TTHTT, $X = 3$). Calculate $\mathbb{E}(X)$ and $\mathrm{Var}(X)$. (Hint: write X as a sum of some indicator variables of some pretty simple events.)

11.3 We throw with a die n times. Let X be the number of times we throw 1, and Y be the number of times we throw 2. What is the correlation of the two random variables?

HW 11.4 (3 points) There are 37 pockets on the roulette wheel, from 0 to 36. Xavier always bets that the result is at least 19. Yvette always bets that the result is 1 mod 3 (so, $1, 4, 7, \ldots, 34$). Let's spin the wheel 20 times, independently. Let X be the number of times Xavier wins, and Y be the number of times Yvette wins. (It's possible that they both win a round, or that neither wins a round). What is the correlation of X and Y?

HW 11.5 (3 points) In an urn, there are 9 pink and 13 orange balls. We take out all balls without replacement. Let X be the number of *pure runs*. (Similarly to question 11.2, but with pink and orange instead of heads and tails). Calculate $\mathbb{E}(X)$.

11.6 Twelve people get on an elevator. They all choose a destination independently out of the 10 possible floors. Determine the expected value and the variance of the number of floors where the elevator stops.

11.7 Let (X, Y) has jointly uniform distribution on the triangle determined by the points $(-1, 0)$, $(0, 0)$, $(0, 2)$. (a) What is the two dimensional covariance matrix of (X, Y)?

(b) Let $Z = 7X + 2Y$. What is the two dimensional covariance matrix of (X, Z)?

HW 11.8 (4 points) Let X and Y be random variables that can only take two values: $X \in \{x_1, x_2\}$ and $Y \in \{y_1, y_2\}$. Prove that if $\mathrm{Cov}(X, Y) = 0$, then they are independent. (Not true in general!)

11.9 Let (X, Y) be the coordinates of a uniformly chosen point on the circumference of the circle centered at $(1, 1)$ with radius 1. $\mathrm{Cov}(X, Y) =?$

Probability 1 – Exercises

Tutorial no. 12 — 4th Dec 2025

12.1 Let X and Y be two absolutely continuous random variables. Let X have a marginal PDF of $f_X(x) = \lambda^2 x \exp(-\lambda x)$ if $x \geq 0$, and 0 otherwise. Conditional on $X = x$, let Y be uniform on the interval $[0, x]$. What is:

(a) the joint PDF of (X, Y)?

(b) the marginal PDF of Y, and $\mathbb{E}(Y)$?

(c) $\mathbb{E}(X \mid Y = y)$ and $\mathbb{E}(Y \mid X = x)$?

(d) $\mathbb{E}(X)$? (Hint: use previous results)

(e) $\mathrm{Cov}(X, Y)$?

12.2 Let $X_1, \ldots, X_n$ be independent and identically distributed random variables (iid).

What is $\mathbb{E}(X_1 \mid X_1 + \cdots + X_n)$?

HW **12.3** (2 points) Let X and Y be i.i.d. (independent and identically distributed) positive random variables.

$$\mathbb{E}\left(\frac{X}{X+Y}\right) = ?$$

12.4 Let Y be a normal variable with mean μ and variance 1, and the conditional distribution of X given $Y = y$ is normal variable with mean y and variance σ^2. What is the conditional distribution of Y given $X = x$?

Interpretation: Y is the original signal, X is the observed noisy version of the signal (i.e., the additive noise was an independent normal variable with mean zero and variance σ^2) and after detecting that value of the noisy version, we want to infer the original signal.

HW **12.5** (3 points) We take a PQ line segment with length 2 and break it into two parts, at a uniformly chosen point R. Let X denote the length of the PR line segment. The other, $2 - X$ long line segment is broken again, at a uniformly chosen point S. Let Y denote the length of the RS line segment.

(a) What is the joint PDF of (X, Y)?

(b) Find $\mathbb{E}(Y \mid X = x)$.

(c) Calculate $\mathrm{Cov}(X, Y)$.

♣ **12.6** (3 points) Let X, Y and Z be independent random variables. Let the CDF of X and Y be $F(x)$ and $G(y)$, respectively. Let Z be a Bernoulli random variable with parameter p. Determine the CDF of the following distributions:

(a) $T := ZX + (1 - Z)Y$,

(b) $U := ZX + (1 - Z)\max\{X, Y\}$,

(b) $U := ZX + (1 - Z)\min\{X, Y\}$.

12.7 We roll a die. If the result is i, then we choose Y uniformly on $[0, i]$. What is $\mathbb{E}(Y)$ and $\mathbb{D}^2(Y)$?

12.8 The amount of time until a lightbulb breaks follows an exponential distribution with 9 months expected time. If the lightbulb breaks, I immediately replace it with the same kind of lightbulb. I use my lamp for T years, where T is a number uniformly chosen between 0 and 9 years. Let Y be the number of times I have to change the lightbulb during this time. $\mathbb{E}(Y) = ?$ and $\mathbb{D}^2(Y) = ?$

12.9 Let U_1 and U_2 be two independent random variables with uniform distribution on $[0,1]$. Show that if

$$X = \sqrt{-2\ln U_1} \cdot \cos(2\pi U_2)$$
$$Y = \sqrt{-2\ln U_1} \cdot \sin(2\pi U_2)$$

Then (X, Y) have a joint Normal distribution.

HW 12.10 (3 points) Let X and Y be the two coordinates of a point that we choose randomly on the disc with radius 1 around the origin. ($f(x,y) = \frac{1}{\pi}$ if $x^2 + y^2 \le 1$.) What is the joint density function of $R = \sqrt{X^2 + Y^2}$ and $\Theta = \mathrm{arctg}(Y/X)$?

12.11 Let $X, Y \sim \mathrm{Exp}(\lambda)$ be independent. What is the joint density of $U := X + Y$ and $V := \frac{X}{X+Y}$? Show that U and V are independent.

HW 12.12 (2 points) Let $X_1, \dots, X_n, X_{n+1}$ be independent, standard normal random variables, and let

$$Y = \frac{X_1 + \cdots + X_n}{n}.$$

(a) What is the covariance matrix of (X_1, Y)?

(b) Find $\mathbb{P}(|Y| \le |X_{n+1}|)$.

12.13 Two rival research groups are trying to estimate the mass of a particle. The first group performs k measurements, while the second group performs l measurements. Both research groups estimate the mass of the particle by taking the empirical average of their respective measurements. The true mass of the particle is μ, and the (noisy) measurements have normal distribution with mean μ and variance of σ^2, independently of each other. What is the probability that the estimate of the first research group is better than that of the second research group?

12.14 Let X be a standard normal distribution, and I independent from X, with $\mathbb{P}(I = 1) = \mathbb{P}(I = 0) = \frac{1}{2}$. Define Y as

$$Y := \begin{cases} X, & \text{if } I = 1 \\ -X, & \text{if } I = 0. \end{cases}$$

(a) Show that Y is standard normal.

(b) Are I and Y independent?

(c) Are X and Y independent?

(d) Show that $\mathrm{Cov}(X, Y) = 0$.